

Multiple Regression II: Categorical Predictors, Model Comparison, Nonlinearity, and Robust Uncertainty

Jinsong Chen

Table of contents

1	Predictors That Are Categories	4
1.1	One indicator, then several	4
1.2	What the indicator coefficients mean	4
1.3	Changing the reference level	6
1.4	One-way analysis of variance as a regression	6
2	Contrasts and Adjusted Group Comparisons	7
2.1	A contrast and its standard error	7
2.2	Testing a block of coefficients	8
2.3	Analysis of covariance	9
2.4	Three analyses of a pretest–posttest study	11
2.5	Parallel slopes and the limits of adjustment	13
3	Comparing Nested Models	14
3.1	Comparing residual sums of squares	14
3.2	The same test in matrix form	15
3.3	Information criteria: AIC and BIC	15
4	Changing a Predictor’s Form or the Outcome’s Scale	16
4.1	Adding a squared term	16
4.2	A log-transformed outcome	18
4.3	Standardized coefficients and unique contributions	19
5	Robust Standard Errors for Unequal Spread	20
5.1	Unequal spread and HC3 standard errors	20
5.2	A robust block test	21
5.3	Limits of robust standard errors	21
6	Resampling the Records	22
6.1	Choosing what to resample	22

6.2	Case resampling for the full model	22
6.3	Bootstrap intervals and their limits	24
7	Clustered Data: Project STAR	25
7.1	The study and its question	25
7.2	Which records enter the analysis	25
7.3	The class-type difference as a regression slope	26
7.4	Adjusted and unadjusted models on the same rows	27
7.5	Clustered standard errors	27
7.6	Reporting the result and its limits	29
8	Reading Bridge	30
	Key terms	31
	Further reading	32
	Exercises	33
	Exercise 1: Read indicator coefficients (Foundational)	33
	Exercise 2: A contrast needs covariance (Foundational)	33
	Exercise 3: A nested comparison from residual sums of squares (Foundational)	33
	Exercise 4: What information criteria compare (Foundational)	34
	Exercise 5: Percent and curvature (Integrative)	34
	Exercise 6: Choose the uncertainty calculation (Integrative)	34
	Exercise 7: One randomized study, three analyses (Integrative)	35
	Appendix A: Matrix Geometry, Restrictions, and Sandwich Constructions	35
	Projection and the normal equations	35
	General linear restrictions	36
	The HC3 sandwich	36
	The CR1 cluster sandwich	37
	What least squares estimates when the mean is not linear	37
	Appendix B: Bootstrap Schemes and Intervals	38
	References	39

Chapter overview

Chapters 7 and 8 fitted lines and additive models and read them as conditional comparisons. Along the way they used tools they could only name: a factor coded as 0/1 columns, a contrast between two categories, a joint test of several added coefficients, an information criterion, a squared predictor, a logged outcome, a covariance estimate that allows unequal spread, and a covariance that respects pupils sitting in schools. This chapter develops each of them once, on the same 98 occupation records, on a small randomized reading study with a pretest, and on Project STAR, a randomized class-size study in which pupils sit in schools. By the end of the chapter you will be able to code and read a factor with several levels, estimate a contrast with the right uncertainty, read an analysis of covariance as a regression on a randomized pretest–posttest study, compare nested models and read AIC and BIC, change a predictor’s form or an outcome’s scale and say what question changed, choose between classical, HC3, cluster-robust, and bootstrap uncertainty for a stated reason, and say what none of these calculations repairs.

A regression report rarely stops at one additive equation, and this chapter takes the usual further steps in order. Section 1 enters a factor as indicator columns and shows that the one-way analysis of variance is that regression. Section 2 compares groups through contrasts and then adjusts a randomized comparison for a pretest, the analysis of covariance. Section 3 compares nested models, and Section 4 changes a predictor’s form or the outcome’s scale. Sections 5 and 6 recalculate the uncertainty when the spread is unequal, by a robust covariance and by resampling, and Section 7 meets a design in which pupils sit in schools, so that the uncertainty has to follow the schools. Each step has a common misreading: taking a coefficient as “the effect of type”, adding two standard errors, comparing criteria across different outcomes, or treating a robust standard error as a repair for a design problem.

The occupation records of Chapter 8 carry most of the work: 98 complete rows, income in thousand 1971 Canadian dollars, and the additive model of education, prestige, percentage of women, and type, with the type levels *bc*, *prof*, and *wc*. Every tool is therefore seen on numbers already read, and every adjusted type comparison holds education, prestige, and the percentage of women fixed. The R code shown is the one call that produces each new quantity; the Computer Lab carries the preparation, the checks, and the variations. Where Chapter 8’s appendices already worked a tool through on these records (the group means recovered by indicators, the overall F , the HC3 covariance, and the Normal likelihood behind AIC and BIC), this chapter points back and adds the new use.

1 Predictors That Are Categories

1.1 One indicator, then several

Chapter 7, Section 3.3, fitted a single 0/1 predictor to 24 invented feedback-by-practice scores. The fitted line passed through the two group means, its slope was their difference, 4 with SE 1.6967 and $t = 2.3575$ on 22 df, and that test was the equal-variance two-sample test written as a regression. The same idea extends to more than two groups.

A factor with J levels enters a regression as $J - 1$ indicator columns. One level is omitted and becomes the **reference**: the intercept is its fitted mean, and each indicator coefficient is a difference from it. With three occupation types and blue-collar as the reference,

$$\hat{y}_g = \hat{\alpha} + \hat{\delta}_{\text{prof}} D_{\text{prof}} + \hat{\delta}_{\text{wc}} D_{\text{wc}}, \quad \hat{\mu}_{\text{bc}} = \hat{\alpha}, \quad \hat{\mu}_{\text{prof}} = \hat{\alpha} + \hat{\delta}_{\text{prof}}, \quad \hat{\mu}_{\text{wc}} = \hat{\alpha} + \hat{\delta}_{\text{wc}}.$$

Two indicators, not three, carry three means. A third column would equal the intercept column minus the other two, so the design matrix would lose full rank and no coefficient would be separately estimable. Which level is omitted is a choice of coordinates, not a choice of model.

The Chapter 8 records and models are re-created here so that every later calculation has a stated origin: `m0`, `m1`, and `m2` are the education-only, the education-and-prestige, and the full additive models.

```
prestige <- carData::Prestige
prestige$income_k <- prestige$income / 1000
d <- prestige[!is.na(prestige$type), ]
d$type <- factor(d$type, levels = c("bc", "prof", "wc"))
m0 <- lm(income_k ~ education, data = d)
m1 <- lm(income_k ~ education + prestige, data = d)
m2 <- lm(income_k ~ education + prestige + women + type, data = d)
```

The 98 records hold 44 blue-collar, 31 professional, and 23 white-collar occupations, and the full model fits six coefficients, leaving 92 residual degrees of freedom. The rows were chosen because their records are complete, not sampled at random from a labour market (Fox et al., 2022; Fox & Weisberg, 2019).

1.2 What the indicator coefficients mean

Fit income on type alone and read the coefficients against Figure 1. A blue-collar occupation carries 0 in both indicator columns, a professional one a 1 in the first, and a white-collar one a 1 in the second, so each type's fitted value is the intercept plus that type's coefficient.

```
group_fit <- lm(income_k ~ type, data = d)
coef(group_fit)
```

(Intercept)	typeprof	typewc
5.374136	5.185315	-0.321832

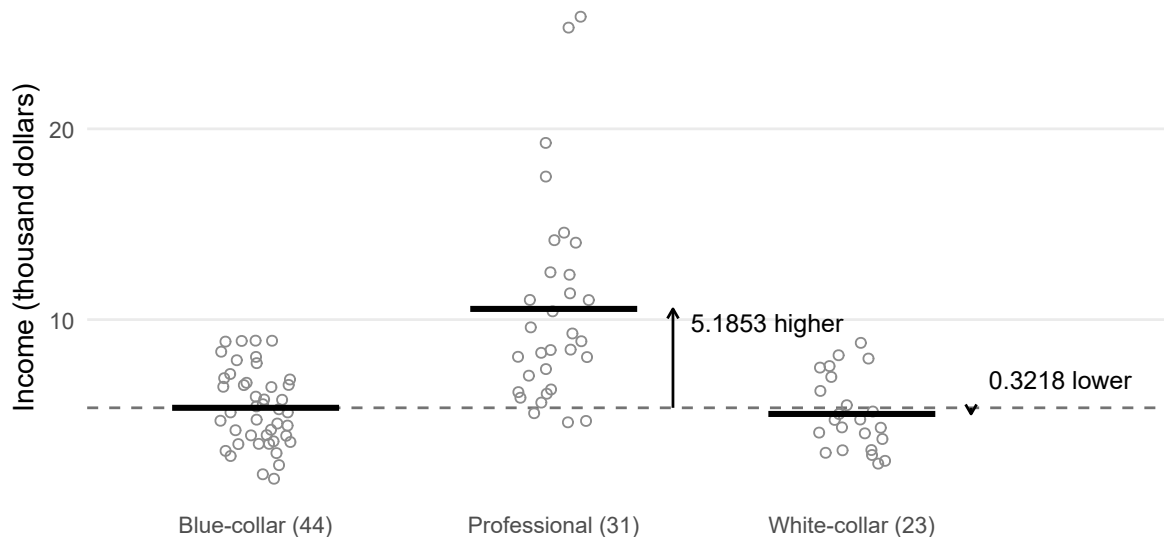


Figure 1: Income of the 98 occupations by type. Bars mark the group means; the dashed line is the blue-collar mean, the intercept, and the arrows are the two indicator coefficients, 5.1853 and -0.3218 .

The coefficients are 5.3741, 5.1853, and -0.3218 . The intercept is the blue-collar mean, the dashed line in the figure, and each coefficient is an arrow from that line to another type's mean: professional occupations average 5.1853 thousand dollars more, white-collar occupations 0.3218 less. The fitted means, 5.3741, 10.5595, and 5.0523, are exactly the observed means, for the reason Chapter 8, Appendix A, gave: with one free parameter per group, least squares returns each group's own mean. A group-only regression is therefore a description of three means, with no adjustment in it. The figure also shows the wide spread of professional incomes, from about 4.6 to 25.9 thousand dollars, which Section 5 takes up.

Keep this description apart from the full model. In `m2` the professional and white-collar coefficients are 0.3242 and 0.2353: they compare fitted incomes at the *same* education, prestige, and percentage of women. The raw professional advantage of 5.1853 and the adjusted 0.3242 answer different questions, and neither is “the effect of occupational type”.

1.3 Changing the reference level

Releveling makes a different level the reference. Refit the group-only model with white-collar as the reference:

```
group_fit_wc <- lm(income_k ~ relevel(type, ref = "wc"), data = d)
coef(group_fit_wc)
```

```
(Intercept)    relevel(type, ref = "wc")bc
      5.052304                0.321832
relevel(type, ref = "wc")prof
      5.507147
```

The coefficients are now 5.0523, 0.3218, and 5.5071: the intercept is the white-collar mean, and the two coefficients measure the blue-collar and professional means from it. The fitted values and residuals are unchanged (Chapter 8, Appendix A); the same three means are reported in new coordinates. Releveling `m2` in the same way gives intercept 0.6559, blue-collar -0.2353 , and professional 0.0889, while the education, prestige, and women coefficients stay exactly as they were.

Centering a continuous predictor, subtracting a constant such as its mean, is the same kind of change: it moves the profile the intercept describes and leaves every fitted value, residual, and substantive contrast alone. Neither operation adds information, such as the cross-type support the records lack, and “the professional coefficient” always means professional minus whichever level is currently the reference.

1.4 One-way analysis of variance as a regression

Chapter 4 tested whether several group means are equal with a one-way analysis of variance. In the group-only regression, equal means says that both indicator coefficients are zero, so the reduced model is the intercept-only model, whose residual sum of squares is the total sum of squares about the grand mean. The test compares the two models’ residual sums of squares:

$$F = \frac{(RSS_{\text{reduced}} - RSS_{\text{full}})/2}{RSS_{\text{full}}/(98 - 3)} = \frac{(1734.0792 - 1138.123)/2}{1138.123/95} = 24.8725.$$

The numerator is the drop in residual sum of squares that the two indicators buy, per indicator; the denominator is the group-only model’s residual mean square on its 95 degrees of freedom.

```
anova(lm(income_k ~ 1, data = d), group_fit)
```

Analysis of Variance Table

Model 1: income_k ~ 1

Model 2: income_k ~ type

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	97	1734.1				
2	95	1138.1	2	595.96	24.872	2.057e-09 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

`aov()` returns the same statistic from the same fitted means, residual sum of squares, and degrees of freedom. The equality is algebraic: it holds whether or not group membership was randomized, so it supplies no causal warrant of its own. As in Chapter 4, the omnibus F says that the three means are not all equal but not which differ, and a comparison chosen after seeing it needs Chapter 4's family calibration. The next section gives the calculation for a single declared comparison.

2 Contrasts and Adjusted Group Comparisons

2.1 A contrast and its standard error

A **contrast** is a linear combination of coefficients, $C = \mathbf{c}^T \boldsymbol{\beta}$, declared before the comparison is read. Its estimate is the same combination of the fitted coefficients, and its variance uses the whole coefficient covariance matrix $\widehat{\mathbf{V}}$:

$$\widehat{C} = \mathbf{c}^T \widehat{\boldsymbol{\beta}}, \quad \widehat{\text{Var}}(\widehat{C}) = \mathbf{c}^T \widehat{\mathbf{V}} \mathbf{c}, \quad SE(\widehat{C}) = \sqrt{\mathbf{c}^T \widehat{\mathbf{V}} \mathbf{c}}.$$

For a difference of two coefficients the variance reduces to $\text{Var}(b_1) + \text{Var}(b_2) - 2 \text{Cov}(b_1, b_2)$. Take professional minus white-collar in `m2`, whose coefficients are ordered intercept, education, prestige, women, professional, white-collar, so that $\mathbf{c} = (0, 0, 0, 0, 1, -1)^T$.

```

c_vec <- c(0, 0, 0, 0, 1, -1)
V <- vcov(m2)
c(estimate = sum(c_vec * coef(m2)),
  SE = sqrt(drop(c_vec %*% V %*% c_vec)))

```

```

      estimate      SE
0.08886278 1.07999307

```

The estimate is $0.3242 - 0.2353 = 0.0889$ thousand dollars. Its variance combines the two coefficient variances and their covariance,

$$2.1406 + 0.9698 - 2(0.972) = 1.1664,$$

so the SE is 1.08 and, with $t_{.975,92} = 1.9861$, the 95% interval is $[-2.0561, 2.2338]$.

The covariance matters because the two type coefficients, both measured from the blue-collar reference, move together: their correlation is .6746, so a sample that overstates one tends to overstate the other, and their difference varies less than it would if the two moved independently. Dropping the covariance would give an SE of 1.7636, and adding the two standard errors, $1.4631 + 0.9848 = 2.4479$, gives a number that is not the standard error of anything.

The vector **c** is the question written down. One entry set to 1 and the rest to 0 returns an ordinary coefficient. Entries on the type coefficients that sum to zero, with the intercept entry 0, make a contrast in Chapter 4's sense, which names the same comparison whichever level is the reference: the relevelled fit of Section 1.3 also gave 0.0889. A contrast chosen because it looked large in the output needs Chapter 4's calibration for the family it was chosen from. Here the interval covers zero, so the records leave the comparison unresolved, and it remains a working-model calculation for types that barely overlap on the other predictors (Chapter 8).

2.2 Testing a block of coefficients

A contrast tests one declared combination. A **block test** asks whether a set of coefficients is jointly zero; it is Section 1.4's comparison of two models with a different reduced model. Two blocks ask different questions here.

```

no_type <- lm(income_k ~ education + prestige + women, data = d)
anova(m1, m2)
anova(no_type, m2)

```


Adding the percentage of women and the two type indicators to `m1` tests three restrictions against the full model's 92 residual degrees of freedom; dropping type from the full model tests two restrictions against the same denominator.

Table 1

Two declared adjusted block comparisons.

Added	df	F	p
Women and type	3	11.5088	< .0001
Type after women	2	0.0319	.9686

The first block is strongly supported: $F(3, 92) = 11.5088$ against a 5% critical value of 2.7036. The second is not: $F(2, 92) = 0.0319$, $p = .9686$, against 3.0954. Joint evidence for the first block does not show that each added column contributes; the percentage of women could carry the whole block. The second result does not show that occupational type is irrelevant: once education, prestige, and the percentage of women are in the equation, the two type indicators reduce the residual sum of squares by almost nothing *on these 98 records*, where the types barely overlap on the other predictors.

A single `anova(m2)` answers a different question. It gives **sequential** sums of squares: each line tests a term given only the terms above it in the formula, so the answers depend on the order of the formula, and only the last line matches a block test. Section 5.2 gives the robust version of the type block.

2.3 Analysis of covariance

The occupation types differ in many ways at once. A randomized study with a baseline score lets the group question be asked more sharply. Baumann and colleagues randomly assigned 68 fourth-grade pupils in one school to three reading-instruction conditions (Baumann et al., 1992): DRA, a directed reading activity from the basal programme; DRTA, a directed reading-thinking activity; and think-aloud strategy instruction. Each pupil took an error-detection test of comprehension monitoring before and after the lessons. `carData::Baumann` holds the 66 pupils with a recorded first posttest, 22 per group. We use the first pretest and posttest, with DRA as the reference; the investigators also analysed further outcomes and both pretests.

The **analysis of covariance** adds the pretest to the group comparison as one more predictor with a single slope:

$$Y_{ij} = \mu_j + \beta(X_{ij} - \bar{X}) + \varepsilon_{ij}, \quad \bar{Y}_j^{\text{adj}} = \bar{Y}_j - \hat{\beta}(\bar{X}_j - \bar{X}).$$

Here Y_{ij} and X_{ij} are the posttest and pretest of pupil i in group j . The group terms μ_j shift the line up or down, and the pretest enters once, so every group's line has the same slope β .

An **adjusted mean** is a group's fitted mean at a stated pretest value. At the grand pretest mean $\bar{X}$ it is the observed posttest mean $\bar{Y}_j$ moved along the common slope by the group's pretest departure from $\bar{X}$, the second equation.

```
reading <- carData::Baumann
reading$group <- factor(reading$group,
  levels = c("Basal", "DRTA", "Strat"),
  labels = c("DRA", "DRTA", "Think-aloud"))
ancova <- lm(post.test.1 ~ group + pretest.1, data = reading)
coef(ancova)
```

(Intercept)	groupDRTA	groupThink-aloud	pretest.1
-0.5966478	3.6265538	2.0361644	0.6931872

```
predict(ancova, interval = "confidence",
  newdata = data.frame(group = levels(reading$group),
    pretest.1 = mean(reading$pretest.1)))
```

	fit	lwr	upr
1	6.188185	5.143481	7.232889
2	9.814739	8.779995	10.849482
3	8.224349	7.181274	9.267425

Read Figure 2 from each filled symbol, a group's observed pretest and posttest means, to the open symbol on the vertical line, its adjusted mean. DRA happened to start highest, 0.7121 points above the grand pretest mean of 9.7879, so following its line to the grand mean lowers its posttest mean by 0.6932×0.7121 , from 6.6818 to 6.1882. Think-aloud started 0.6515 points below and rises from 7.7727 to 8.2243, while DRTA, almost at the grand mean, barely moves, from 9.7727 to 9.8147. The fit gives the common slope 0.6932 (SE 0.1015), differences from DRA of 3.6266 (SE 0.7362) for DRTA and 2.0362 (SE 0.7450) for Think-aloud, and a residual mean square of 5.894 on 62 degrees of freedom. Tested together after the pretest, the two group coefficients give $F(2, 62) = 12.1689$, $p < .0001$; the adjusted means have 95% intervals [5.1435, 7.2329], [8.7800, 10.8495], and [7.1813, 9.2674].

Three conditions carry that reading. The adjusted means need a **common reference**, here the grand pretest mean. The reference must lie inside every group's observed pretest range, the **common support**: the pretests run from 4 to 16, 6 to 16, and 4 to 14, so the lines are drawn over 6 to 14, and a comparison read outside that band would rest on extrapolation. And the comparison needs a stated **averaging population**: for this additive model, predicting every group at the grand pretest mean gives the same values as predicting every group at each of the 66 pupils' pretests and averaging. Because the lines are parallel, the adjusted differences, 3.6266 and 2.0362, are the same at any pretest value inside the common support; the adjusted means are not, which is why the reference value is stated.

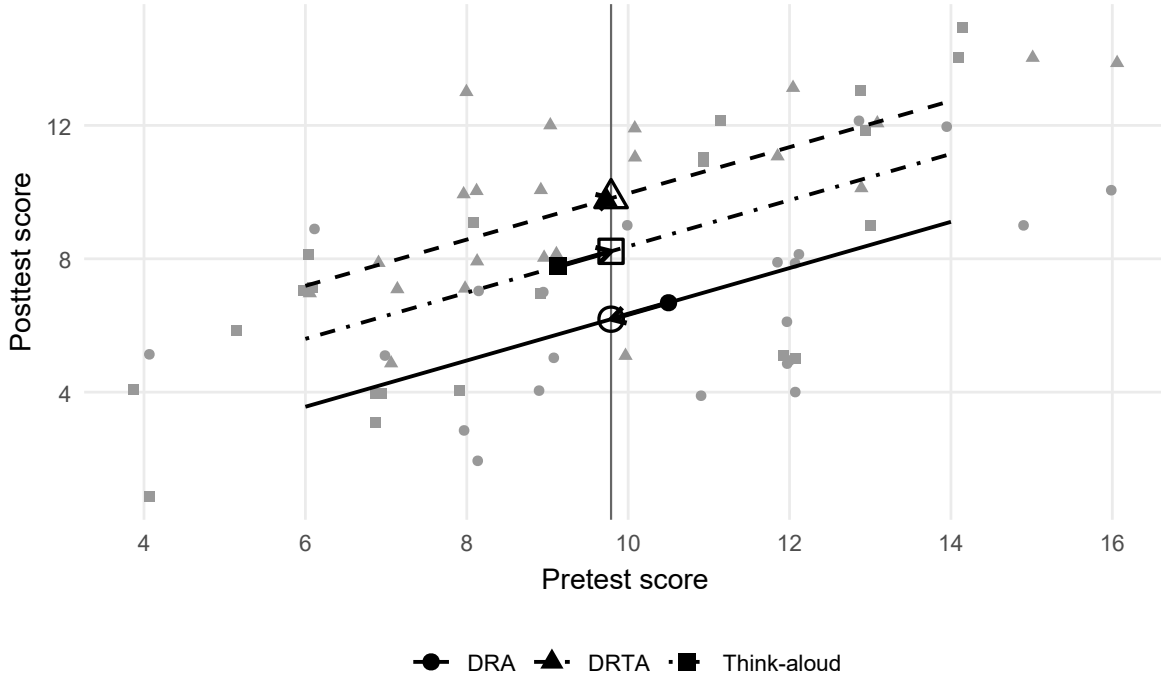


Figure 2: Posttest against pretest for the 66 Baumann pupils. The parallel lines are the common-slope fit over the common pretest support 6–14; each filled symbol marks a group’s observed pretest and posttest means, and its arrow follows the group’s line to the grand pretest mean 9.7879, where the open symbol marks the adjusted mean.

2.4 Three analyses of a pretest–posttest study

A randomized pretest–posttest study is often analysed in one of three ways: compare the posttest means, compare the gains (posttest minus pretest), or adjust for the pretest as above. The three are one regression that differs only in the pretest coefficient γ :

$$Y_i - \gamma X_i = \alpha + \delta_{\text{DRTA}} D_{\text{DRTA},i} + \delta_{\text{TA}} D_{\text{TA},i} + \varepsilon_i, \quad \gamma = 0, 1, \text{ or estimated.}$$

With $\gamma = 0$ the model compares posttest means, with $\gamma = 1$ it compares gains, and with γ estimated it is the analysis of covariance. In R, `offset()` enters the pretest with its coefficient fixed at 1:

```
post_only <- lm(post.test.1 ~ group, data = reading)
gain <- lm(post.test.1 ~ group + offset(pretest.1), data = reading)
```

Under random assignment the expected pretest difference between groups is zero, so all three analyses estimate the same average difference in posttest scores. Random assignment covered the 68 pupils assigned, while the analyses use the 66 with a recorded first posttest, so the argument also needs a complete-record condition, for example that the two posttests went missing for reasons unrelated to what those pupils would have scored, which the design does not guarantee and the records cannot check.

Table 2

Three analyses of the Baumann records as one regression.

Analysis	Pretest coefficient	DRTA minus DRA (SE)	Think-aloud minus DRA (SE)	Residual SD (df)
Posttest only	0	3.0909 (0.9614)	1.0909 (0.9614)	3.1885 (63)
Gain score	1	3.8636 (0.7779)	2.4545 (0.7779)	2.5799 (63)
Analysis of covariance	0.6932, estimated	3.6266 (0.7362)	2.0362 (0.7450)	2.4278 (62)

The three analyses differ in precision and in how this sample's chance pretest imbalance enters. The pretest means are 10.5000, 9.7273, and 9.1364, so DRA had a head start of 0.7727 points over DRTA, about a quarter of the pretest SD of 3.0205 across all 66 pupils. The size of that gap is what matters. A significance test of it would add nothing: random assignment already makes any baseline gap a chance gap, so the test would examine a null hypothesis known to be true. The gap accounts exactly for the spread of the estimates in Table 2: each DRTA estimate is the posttest-only difference plus γ times DRA's head start, $3.0909 + \gamma(10.5 - 9.7273)$, which gives 3.0909, 3.8636, and 3.6266 (3.6265 from the rounded inputs).

Estimating the coefficient leaves the smallest residual SD, 2.4278 against 3.1885 and 2.5799, and the smallest standard errors, because the pretest predicts the posttest with a slope near 0.7 rather than 0 or 1. The choice belongs in the analysis plan, made before the outcomes are seen, not after comparing the three estimates.

2.5 Parallel slopes and the limits of adjustment

A common slope is a condition, not a finding. The check adds the group-by-pretest product block, which lets each group have its own slope, and compares the two named models on the same records:

```
varying_slopes <- lm(post.test.1 ~ group * pretest.1, data = reading)
anova(ancova, varying_slopes)
```

Analysis of Variance Table

```
Model 1: post.test.1 ~ group + pretest.1
Model 2: post.test.1 ~ group * pretest.1
  Res.Df    RSS Df Sum of Sq    F Pr(>F)
1      62 365.43
2      60 347.97  2    17.461 1.5054 0.2302
```

The product block gives $F(2, 60) = 1.5054$, $p = .2302$, with group slopes of 0.4663 (DRA), 0.6868 (DRTA), and 0.8768 (Think-aloud). These 66 records do not distinguish the two models, which is not the same as showing that the slopes are equal. When slopes do differ, an adjusted difference depends on the pretest value at which it is read; Chapter 10 develops that varying-slope model as an interaction.

What adjustment supplies depends on how the groups were formed. In groups formed by random assignment, the pretest only sharpens a comparison that the design already warrants. In observational groups, a fitted comparison at a common covariate value cannot balance what was not measured; a covariate measured with error under-adjusts, leaving part of the baseline difference in place (Chapter 14, Section 5.1); and a covariate recorded after the groups formed can remove part of the effect being estimated.

Lord's paradox makes the point with intact groups that differ at baseline (Lord, 1967). The gain-score analysis asks whether the groups changed by different amounts; the covariance analysis asks whether pupils with the same pretest ended differently. The two can reach opposite conclusions from the same records, partly through the regression toward the mean of Chapter 7, Section 7. Neither analysis says which question is the causal one; the design does, and Chapter 14 returns to what a design supplies that adjustment cannot.

3 Comparing Nested Models

3.1 Comparing residual sums of squares

Sections 1.4 and 2.2 used one idea. Two models are **nested** when the reduced model is the fuller model with some coefficients set to zero, and both are fitted to the same records and outcome. Extra coefficients never raise the residual sum of squares; the question is whether they lower it by more than noise alone would. The **extra-sum-of-squares** statistic compares the two residual sums of squares:

$$F = \frac{(RSS_R - RSS_F)/q}{RSS_F/(n - p)}.$$

The numerator is the drop that the q extra coefficients buy, per coefficient. The denominator is the fuller model's residual mean square, with p its number of coefficients: the same s^2 that appears in every classical standard error. Under the Normal linear model with independent common-variance errors, the ratio has an F reference with q and $n - p$ degrees of freedom.

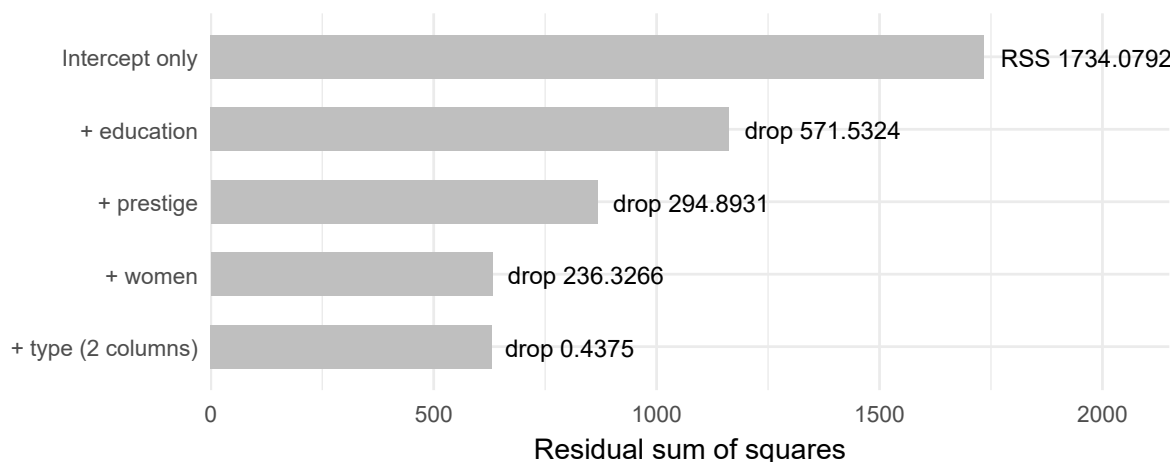


Figure 3: Residual sum of squares as columns are added to the occupation model on the same 98 records; each label gives the drop that the added column or block buys.

Figure 3 shows what each addition buys in the occupation models. Education buys 571.5324 of the total 1734.0792, and prestige a further 294.8931. The percentage of women then buys 236.3266 for one coefficient, while the two type columns buy only 0.4375 for two. Against the full model's residual mean square of 6.8575, the type block's 0.4375 over two coefficients gives $F(2, 92) = 0.0319$, and the women-and-type block's 236.764 over three gives 11.5088, the entries of Table 1. The subtraction means something only when both models use the same

rows and the same outcome, which is why every comparison in this chapter runs on the same 98 records.

The **overall** regression test is the special case whose reduced model is the intercept alone, with all $p - 1$ non-intercept coefficients set to zero. Chapter 8, Appendix A, gave its formula and read it from `summary()`: $F(5, 92) = 32.1747$ for the full model and $F(1, 96) = 47.1956$ for the education-only model on the same records. The larger value is not evidence that `m0` is better, because the two statistics test different restrictions. Neither checks the mean function, and neither says how a fitted rule would predict new records (Chapter 13).

3.2 The same test in matrix form

The statistic also has a compact matrix form. Let $\mathbf{X}$ hold n rows and p coefficient columns including the intercept, with full column rank. Least squares solves the normal equations $\mathbf{X}^\top(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}) = \mathbf{0}$, so that $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{y}$, $\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}$ with $\mathbf{H} = \mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top$, and $\hat{\mathbf{V}} = s^2(\mathbf{X}^\top\mathbf{X})^{-1}$. For q independent linear restrictions $\mathbf{R}\boldsymbol{\beta} = \mathbf{r}$,

$$F = \frac{(\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r})^\top [\mathbf{R}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{R}^\top]^{-1} (\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r})}{q s^2} \sim F_{q, n-p}.$$

For `m2`, $\mathbf{X}$ has six columns and rank 6: the two type indicators do not reproduce the intercept column, so every coefficient is separately estimable. Choosing the two rows of $\mathbf{R}$ that select the type coefficients, with $\mathbf{r} = \mathbf{0}$, returns 0.0319, the type block test of Section 2.2, without fitting a second model.

The equation explains what the estimator *is*, not how R computes it: `lm()` uses a QR decomposition, which is numerically better behaved than forming $(\mathbf{X}^\top\mathbf{X})^{-1}$ (Fox, 2016; Fox & Weisberg, 2019). Because $\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}$ is a projection onto the column space of $\mathbf{X}$, any full-rank recoding of a factor spans the same space and leaves the fitted values alone: Section 1.3's releveling result, stated geometrically. Appendix A carries the derivation.

3.3 Information criteria: AIC and BIC

A nested F test compares two models when one contains the other. Information criteria can also rank models that are not nested, using a different quantity: the maximized likelihood, penalized for the number of fitted parameters.

Chapter 8, Appendix B, built that likelihood for these models by carrying Chapter 6's binomial likelihood for 32 successes in 50 trials over to independent Normal outcomes around the regression. Maximizing it over the coefficients reproduces least squares. Its scale estimate divides the residual sum of squares by n , $\sqrt{RSS/n} = 2.5373$, where the residual SD reported by `lm()` divides by the residual degrees of freedom, $\sqrt{RSS/(n-p)} = 2.6187$. AIC adds $2k$ to

$-2\log \widehat{L}$ and BIC adds $k \log n$, where k counts the fitted coefficients **and** the residual variance, so `m2` has $k = 7$, not 6. One call returns all three quantities:

```
c(logLik = logLik(m2), AIC = AIC(m2), BIC = BIC(m2))
```

logLik	AIC	BIC
-230.3020	474.6040	492.6988

For `m2` it returns -230.3020 , 474.6040 , and 492.6988 . Chapter 8, Appendix B, set the criteria for the three nested models side by side: AIC 526.5052, 499.8328, and 474.6040 for `m0`, `m1`, and `m2`, and BIC 534.2601, 510.1727, and 492.6988. Smaller is preferred, and both criteria prefer `m2`.

The two criteria charge differently for parameters. For the seven parameters of `m2`, BIC charges $7 \log 98 = 32.0948$ instead of AIC's 14, because at $n = 98$ the factor $\log n = 4.585$ exceeds 2, so the criteria can disagree; a disagreement shows how strongly the extra parameters are charged, not a tie to break by preference. The criteria are comparable only among models fitted to the same records, with the same outcome on the same scale, under one likelihood convention; Section 4.2 shows a change of scale breaking the comparison. Both are calculated on the records that fitted the models, so neither is a held-out error, and both are comparative: a smaller AIC says one candidate is preferred to another, not that either describes the records well (Akaike, 1974; Schwarz, 1978). Chapter 11 develops likelihood fully.

4 Changing a Predictor's Form or the Outcome's Scale

4.1 Adding a squared term

In every model so far, a continuous predictor's slope was the same at every value of that predictor. Chapter 8, Section 4.3, refitted the full model with a squared prestige term, and that refit is the worked example here. It was a check on functional form declared before it was run, not a search across candidate terms for the smallest p -value; a p -value read after such a search would not mean what it appears to mean. A squared term is one kind of **polynomial term**, a power of a predictor entered as an ordinary extra column, here `I, and the model is still linear in its coefficients. What changes is the reading:`

$$E(Y | x, \dots) = \beta_0 + \beta_1 x + \beta_2 x^2 + \dots, \quad \frac{\partial E(Y)}{\partial x} = \beta_1 + 2\beta_2 x, \quad x^* = -\frac{\beta_1}{2\beta_2}.$$

The predictor's slope now depends on its own value, $\beta_1 + 2\beta_2 x$, and is zero at the turning point x^* .


```
m2q <- lm(income_k ~ education + prestige + I(prestige^2) + women + type,
  data = d)
anova(m2, m2q)
```

Analysis of Variance Table

Model 1: income_k ~ education + prestige + women + type

Model 2: income_k ~ education + prestige + I(prestige^2) + women + type

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	92	630.89				
2	91	587.41	1	43.481	6.736	0.01101 *

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

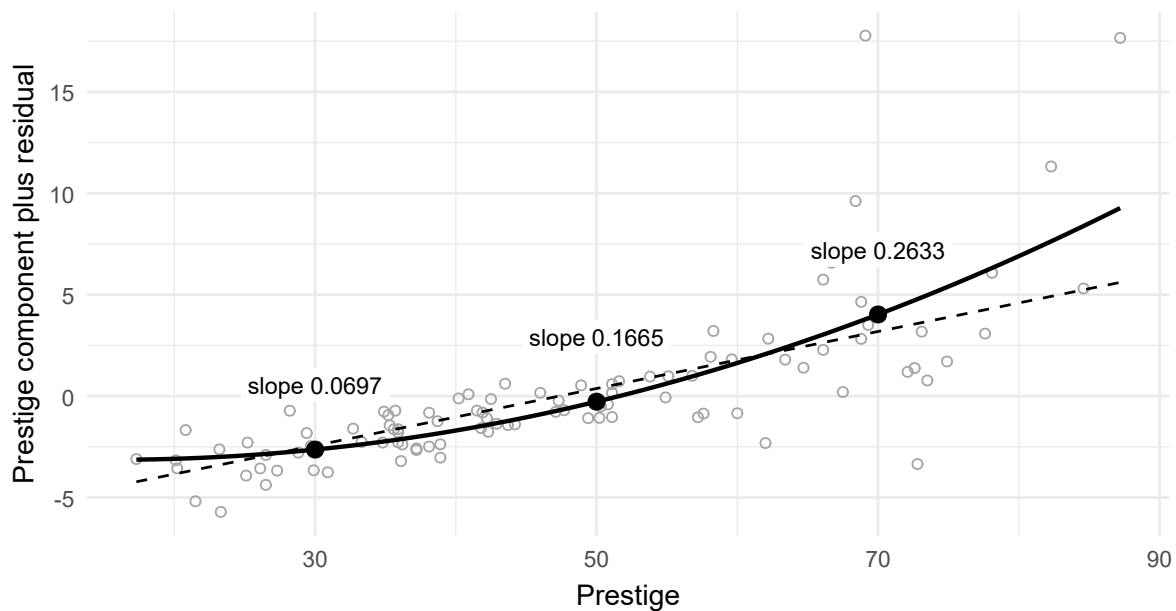


Figure 4: Component-plus-residual display for prestige in the model with a squared prestige term. The solid curve is the fitted quadratic component and the dashed line the linear component of the additive model; the labels give the curve's slope at prestige 30, 50 and 70.

The quadratic coefficient is 0.0024 with SE 0.0009 and $t = 2.5954$; the one-restriction block test gives $F(1, 91) = 6.736$ with $p = .011$; and the criteria move from 474.6040 and 492.6988 to 469.6058 and 490.2855. The education coefficient changes from 0.1164 to -0.0298 , a reminder that each coefficient is defined relative to the other columns present.

Figure 4, Chapter 8’s component-plus-residual display with the fitted curve added, shows what the squared term claims. With linear coefficient -0.0754 and quadratic coefficient 0.0024 , the curve’s slope is 0.0697 thousand dollars per prestige point at prestige 30, 0.1665 at 50, and 0.2633 at 70: income rises with prestige throughout the records, and faster at the top, where the straight line falls below several high-prestige occupations. The turning point, 15.5906 , lies below the smallest observed prestige of 17.3 , so the falling branch of the parabola is outside the records. The coordinate -0.0754 is the slope at prestige zero, a value no occupation has, so it is not a “negative prestige effect”.

Centering fixes that reading without changing the curve, because it is a change of coordinates. Prestige and its square correlate $.9849$; centering prestige at its mean, 47.3276 , before squaring lowers that correlation to $.3216$. The quadratic coefficient and its SE stay 0.0024 and 0.0009 , and every fitted value stays the same. The linear coefficient becomes 0.1536 with SE 0.0347 : the slope at mean prestige, a different quantity from the slope at prestige zero, whose SE is about 2.6 times as large. The smaller SE belongs to a slope read in the middle of the records, not to a sharper estimate of the old one.

4.2 A log-transformed outcome

Changing a predictor’s form keeps the question. Changing the outcome’s scale changes it.

$$\log Y_i = \beta_0 + \sum_j \beta_j x_{ij} + \varepsilon_i, \quad 100(e^{\beta_j} - 1) \text{ percent per unit of } x_j.$$

A coefficient in a log-income model is an additive difference in expected log income. Exponentiating turns it into a multiplicative difference in geometric-mean income under that model, and $100(e^{\hat{\beta}} - 1)$ expresses the multiplier as a percentage. The geometric mean is not the arithmetic mean, so this is not automatically a statement about average income in dollars.

```
m_log <- lm(log(income) ~ education + prestige + women + type, data = d)
100 * (exp(coef(m_log)[c("education", "prestige", "women")]) - 1)
```

```
education  prestige    women
0.7660700  2.0964846 -0.8429719
```

Table 3*Log-model contrasts on a geometric-mean percent scale.*

Term	Log coefficient	Percent	Lower	Upper
education	0.0076	0.7661	−4.2041	5.9941
prestige	0.0207	2.0965	1.4351	2.7622
women	−0.0085	−0.8430	−1.0189	−0.6667

One more prestige point is associated with about 2.0965 percent higher geometric-mean income at the same education, percentage of women, and type, with interval [1.4351, 2.7622] percent; one more percentage point of women with about −0.843 percent. The two type coefficients, −0.0137 and 0.0708 on the log scale, are −1.3584 and 7.3341 percent. The intervals in Table 3 are the classical log-scale intervals with both endpoints transformed, which is legitimate because the transformation is monotone.

The criteria do not carry across this change. This model’s AIC is 8.091 against 474.604 for `m2`, and the difference means nothing: the two likelihoods are densities of different response variables. Keep the outcome and the question fixed inside any comparison, and treat a change of scale as a new question rather than as data cleaning (Fox & Weisberg, 2019).

4.3 Standardized coefficients and unique contributions

Two summaries are often requested once several predictors are in an equation:

$$\beta_j^* = \hat{\beta}_j \frac{s_{x_j}}{s_y}, \quad \Delta R_j^2 = R_{\text{full}}^2 - R_{\text{without } j}^2.$$

A **standardized coefficient**, or beta weight, rescales a partial slope into outcome-SD units per predictor SD. A **drop-one** R^2 reports how much of the explained variation disappears when one predictor is removed from the full model. The beta weights are 0.0757, 0.5685, and −0.3971 for education, prestige, and the percentage of women, and education’s drop-one R^2 is .0007: removing it costs almost nothing in fit, the same message as the education coefficient’s wide interval in Chapter 8.

Neither number measures importance. A standardized coefficient depends on the predictor’s spread in these particular records, so a restricted range shrinks it; it also differs from Chapter 8’s standardization over named profiles, which averages fitted values over a declared distribution. A drop-one change describes only the *unique* contribution: because predictor information overlaps, the separate values need not sum to the joint R^2 , and a predictor that duplicates another can show a near-zero value while mattering. Neither establishes a causal ranking or predicts held-out performance.

5 Robust Standard Errors for Unequal Spread

5.1 Unequal spread and HC3 standard errors

Every classical standard error so far assumed one common error variance for all 98 occupations. Figure 5 shows why that assumption is doubtful here.

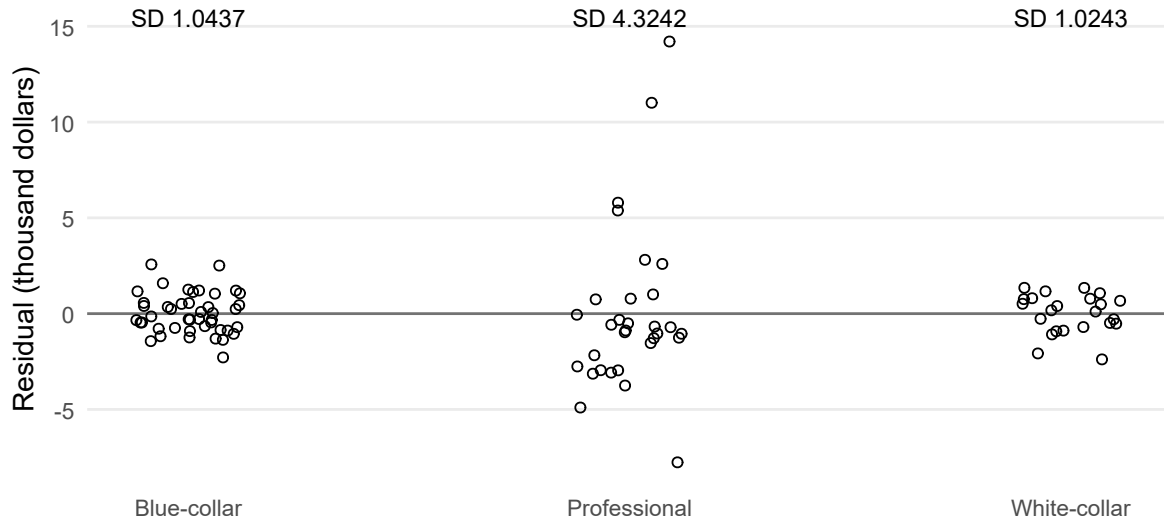


Figure 5: Residuals of the full occupation model by type, with each type's residual SD.

The residuals of professional occupations spread about four times as widely as the others: their SD is 4.3242 thousand dollars against 1.0437 for blue-collar and 1.0243 for white-collar occupations. A single s^2 averages these spreads, so it overstates the uncertainty contributed by some rows and understates it for others.

The remedy is already in hand. Chapter 7, Section 6.1, first used a robust standard error on the simple line (slope SE 0.1270 to 0.1491), and Chapter 8 applied the same HC3 estimator to the full model: its Section 3.3 made the `sandwich::vcovHC()` call, and its Appendix B gave the estimator, which keeps the design component outside and replaces the single s^2 inside by each row's own leverage-adjusted squared residual, with classical and HC3 standard errors for all six coefficients (Long & Ervin, 2000; MacKinnon & White, 1985; White, 1980; Zeileis, 2004). Appendix A of this chapter shows where the estimator comes from.

The figure explains the pattern in those standard errors. The professional coefficient's SE grows by about 40%, from 1.4631 to 2.0928, as the wide professional spread predicts, while the white-collar SE grows by only about 7%, from 0.9848 to 1.0548. The education SE grows from 0.2752 to 0.3823, with robust 95% interval $[-0.6429, 0.8758]$: wider, with the same reading of an imprecise comparison. The SE for the percentage of women *falls*, from 0.0097 to 0.0079, so

a robust covariance is not a penalty applied to every coefficient; it is a different estimate of the same target. Every coefficient, fitted value, and residual stays exactly what it was, because HC3 replaces a covariance estimate, not a fit. Switching estimators after seeing which gives the preferred conclusion turns a stated method into a selection.

5.2 A robust block test

The restriction statistic of Section 3.2 needs only a coefficient vector and a covariance matrix. Substituting $\widehat{\mathbf{V}}_{\text{HC3}}$ for $\widehat{\mathbf{V}}$ gives an approximate Wald version of the same block test: for the type block, 0.0345 with $p = .9661$, against the classical 0.0319 and .9686. Here the substitution changes nothing of substance, because the type coefficients sit close to the restriction, each within a quarter of its classical SE of zero, so both statistics are near zero and far below the .05 critical value, 3.0954. A change of covariance matters most when a statistic lies near its critical value. The two also differ in kind: the classical statistic has an exact $F_{2,92}$ reference under the Normal linear model, while the robust one has only an approximate reference, justified as the sample grows, not the exact finite-sample result of Section 2.2's test.

5.3 Limits of robust standard errors

HC3 has one job: under a linear conditional mean and independent observations, it estimates the coefficient covariance without assuming a common error variance. Several problems lie outside that job.

- **Curvature.** If the mean function is wrong, the coefficients estimate a different quantity (Appendix A names it), and a robust covariance describes that quantity's variability without making it the quantity of interest.
- **Dependence.** Rows that are not independent break the estimator's own derivation. Section 7 shows a robust row-level SE and a naive one agreeing closely while both understate the uncertainty by nearly half.
- **Selection and causal identification.** Selection and missing outcomes change what the records represent, and causal identification comes from the design or from an argument about the conditioning set.
- **Measurement error.** Error in measuring a predictor pulls its slope toward zero, by roughly the predictor's reliability when it is the only predictor, and no covariance estimator repairs it (Cohen et al., 2003); independent mean-zero error in the outcome, by contrast, widens the residual spread and the standard errors but leaves the slope's target unchanged. Chapter 14, Section 5.1, develops attenuation.
- **Thin support.** The type contrast's robust SE of 1.1827 is still the uncertainty of a comparison the records barely observe.

A report therefore names the covariance estimator and the reason for it, and keeps these limits attached to the answer.

6 Resampling the Records

6.1 Choosing what to resample

Every uncertainty statement so far came from a formula: a coefficient covariance matrix, classical or robust, propagated through a linear combination. The **bootstrap** offers a different approximation: refit the model to many resamples of the records and read the spread of the refitted estimates directly (Davison & Hinkley, 1997; Efron, 1979; Efron & Tibshirani, 1993). What to resample is an assumption, not a default.

- **Case resampling** draws whole records with replacement and refits. It inherits whatever unequal spread the rows contain, but it requires the resampling unit to match the sampling unit, with units independent and exchangeable.
- **Residual resampling** holds the design fixed, resamples rescaled residuals, and adds them back to the fitted values. It assumes identically distributed errors, the assumption HC3 exists to relax.
- **The wild bootstrap** also holds the design fixed but multiplies each rescaled residual by an independent mean-zero, unit-variance draw, so every record keeps its own error scale.

This chapter uses case resampling, for two reasons. The occupation predictors were not designed: no one assigned education or prestige to an occupation, so treating the rows as jointly sampled is the natural view. And the residual spread differs by type (Figure 5), so the identically distributed errors behind residual resampling are not available (Freedman, 1981). Appendix B compares the other two schemes on the same targets.

6.2 Case resampling for the full model

Resample the 98 rows with replacement, refit `m2`, and keep the education slope. The seed and the number of resamples B belong in the record, so both appear in the code; B counts resamples here as it counted simulation repetitions in Chapter 5.

```

set.seed(90909)
B <- 2000
boot_education <- replicate(B, {
  rows <- sample.int(nrow(d), replace = TRUE)
  coef(lm(income_k ~ education + prestige + women + type,
    data = d[rows, ]))["education"]
})
c(SE = sd(boot_education), quantile(boot_education, c(.025, .975)))

```

```

      SE      2.5%      97.5%
0.3698895 -0.6829568  0.7439809

```

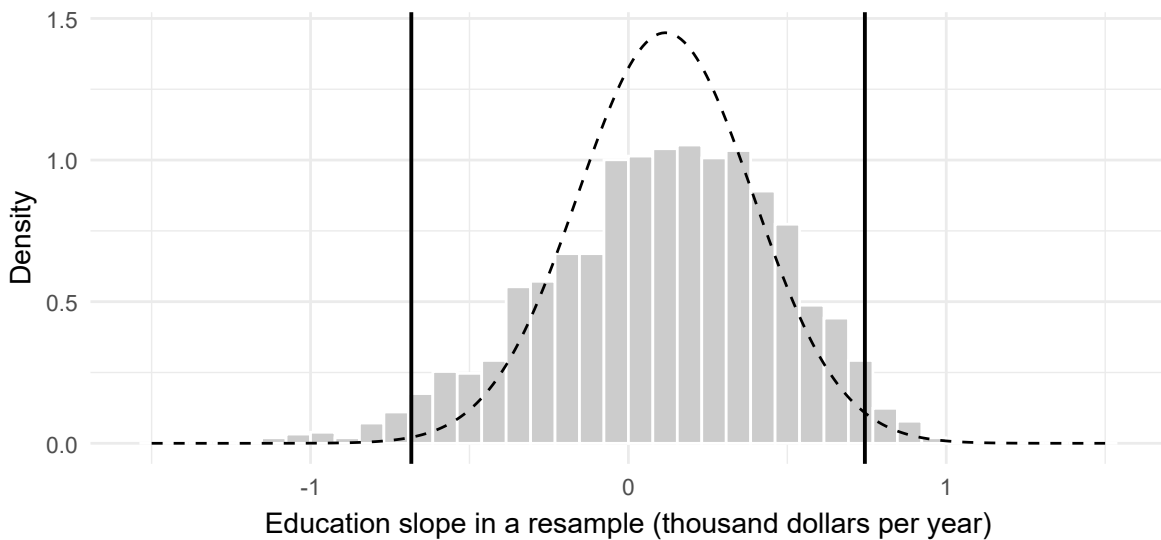


Figure 6: The 2,000 case-resampled education slopes (seed 90909). Solid lines mark the percentile interval; the dashed curve is the Normal distribution with the classical SE 0.2752.

Figure 6 is the resampling distribution itself. The 2,000 refitted slopes spread more widely than the Normal curve built from the classical SE: their SD, the bootstrap SE, is 0.3699, and their middle 95%, the percentile interval, runs from -0.683 to 0.744 . The same resamples give the other two targets of Table 4, the professional-minus-white-collar contrast and the professional coefficient. A resample could omit a whole occupational type and lose a type coefficient; all 2,000 resamples here contained all three types, so all 2,000 refits are usable.

Table 4

Model-based, HC3, and case-bootstrap uncertainty for three targets in the full occupation model (seed 90909, $B = 2000$).

Target	Estimate	Classical SE	HC3 SE	Bootstrap SE	Percentile 2.5%	Percentile 97.5%
Education slope	0.1164	0.2752	0.3823	0.3699	−0.6830	0.7440
prof — wc	0.0889	1.0800	1.1827	1.1187	−1.9613	2.4313
prof coefficient	0.3242	1.4631	2.0928	1.9926	−3.1416	4.5443

Read the table one row at a time. The education slope’s bootstrap SE of 0.3699 sits between the classical 0.2752 and the HC3 0.3823: three calculations, one conclusion that this comparison is imprecise. The type contrast’s bootstrap SE of 1.1187 exceeds the classical 1.0800, with percentile interval $[-1.9613, 2.4313]$. The professional coefficient moves most, from a classical 1.4631 to 1.9926, close to the HC3 2.0928: with only 31 professional occupations spreading as widely as Figure 5 shows, a resample’s composition matters.

Agreement between HC3 and the bootstrap is not independent confirmation: both work on the same 98 records. And a different seed would move the last decimal of a bootstrap summary, which is why the seed and $B = 2000$ are reported.

6.3 Bootstrap intervals and their limits

Two intervals come directly from the resampling distribution. The **percentile interval** takes its 2.5th and 97.5th percentiles, as in Figure 6. The **normal-approximation interval** takes the original estimate plus and minus 1.96 bootstrap standard errors. The two agree when the resampling distribution is roughly symmetric. Here the three distributions have skewness -0.3891 , 0.2635 , and 0.3785 , read by their sign as in Chapter 2, Section 2.1: the education slopes have the slightly longer left tail visible in the figure, the other two a slightly longer right tail, and the asymmetry is mild enough that the two kinds of endpoint agree closely. For a strongly skewed or biased resampling distribution, **bootstrap- t** and **BCa** intervals are the higher-order alternatives; Appendix B defines both in words.

What resampling assumes matters more than which interval is chosen. Drawing occupations with replacement treats them as independent and exchangeable, and there are reasons to doubt both: the occupations share one historical labour market, where common conditions can move related incomes together; the 98 rows were chosen by complete records rather than sampled at random; and their measurements are aggregates over workers. Resampling does not repair thin cross-type support, correct the measurement problems Chapter 8 documented, or make an observational comparison causal. And when a design has clusters, strata, or weights, the resampling unit has to follow the design rather than the rows, which is where the next section begins.

7 Clustered Data: Project STAR

The occupation records were observational, and the reading study was a small experiment in one school. Project STAR is a large experiment of a different shape: kindergarten pupils were randomized to class types within their schools. The randomization supports reading the class-type comparison as an assignment comparison, and the schools make pupils' outcomes dependent, which the uncertainty calculation has to follow.

7.1 The study and its question

The question is specific:

Among eligible kindergarten pupils in the participating Tennessee schools, what was the mean reading-score difference under assignment to a small rather than a regular class?

The outcome is `readk`, an end-of-kindergarten reading scale score, and the field `stark` records the kindergarten class-type condition. Reading the small-versus-regular comparison as an assignment comparison rests on the documented protocol, not on the column name or the `lm()` call: random assignment occurred **within** participating schools (Finn & Achilles, 1990; Mosteller, 1995). The regular-with-aide arm lies outside this two-arm comparison and is excluded by design.

Three limits come with the question. Class **type** is not a measured headcount, so the analysis estimates no dose response in pupils per room. The distributed field does not verify each pupil's original assignment; an assignment-effect reading also requires linking the recorded type to the assignment record and checking switches, which this teaching reanalysis does not do. And the target is the pupil-weighted mean assignment effect for the eligible cohort, which a common coefficient from a complete-record model reaches only through a further link: a common effect across schools or an explicit weighting argument.

The data are distributed as `AER::STAR`, accompanying *Applied Econometrics with R* and following the version used by Stock and Watson (Kleiber & Zeileis, 2008; Stock & Watson, 2007). They are loaded from the package, and the analysis is retrospective rather than preregistered.

7.2 Which records enter the analysis

A record can be absent from an analysis for two different reasons, and they are kept apart. Pupils who entered the study after kindergarten have no kindergarten condition: their absence is **structural ineligibility**. Eligible kindergarten pupils whose reading score or covariates were not recorded are **missing**, and missingness can distort a comparison. Only the variables

a calculation uses enter its complete-record rule, so a missing mathematics score (`mathk`) does not exclude a usable reading record.

Table 5

STAR eligibility and reading-analysis completeness.

Stage	<i>n</i>
Distributed records	11,598
Kindergarten condition recorded	6,325
Eligible small or regular	4,094
Eligible, reading observed	3,745
Complete primary records	3,735
Primary regular	2,001
Primary small	1,734
Primary schools	79

Of 11,598 distributed records, 5,273 have no kindergarten condition; they are later entrants, not failed measurements. The remaining 6,325 have a recorded kindergarten arm, and across all three arms 536 lack a reading score. The two declared arms hold 4,094 eligible records, of which 349 lack reading, 188 in the regular arm and 161 in the small arm, leaving 3,745 with an observed reading score, 2,006 regular and 1,739 small.

Similar missing percentages in the two arms are reassuring but prove nothing, since the absent scores could differ systematically within either arm. Complete-record inference assumes that, given the declared covariates and within arm and school, whether a score was recorded does not depend on the score itself; the assumption is stated here, not verified. Among the observed-reading records across all arms, 38 lack at least one of lunch status, gender, ethnicity, and teacher experience; within the two focal arms, ten do. The primary sample therefore has 3,735 pupils, 2,001 regular and 1,734 small, in 79 schools, and the 3,745 observed-reading records serve as a declared sensitivity analysis.

7.3 The class-type difference as a regression slope

Section 1.1's identity applies unchanged. With $D_i = 1$ for the small arm and 0 for regular,

$$\widehat{Y}_i = \hat{\alpha} + \hat{\delta}D_i, \quad \hat{\alpha} = \bar{Y}_{D=0}, \quad \hat{\delta} = \bar{Y}_{D=1} - \bar{Y}_{D=0} = 440.5474 - 434.7323 = 5.8151.$$

On the 3,745 observed-reading records the slope reproduces the arm difference of 5.8151 reading points exactly, as Chapter 7's binary predictor did for 24 pupils. The identity is algebraic; randomization adds the reason to read the difference as an assignment comparison.

Two cautions attach to the number. Treating the rows as independent with a common variance, the slope's t test equals the pooled two-sample test, here $t = 5.6039$, which ignores that pupils sit in schools (Section 7.5). And raw arm means weight each school by its own arm-specific pupil count, so the pooled difference equals the pupil-weighted cohort target only under common assignment probabilities and comparable observation patterns, or after explicit standardization to target school weights.

7.4 Adjusted and unadjusted models on the same rows

The adjusted model adds free-lunch status, gender, ethnicity, teacher experience, and school indicators. The school indicators match the blocks within which assignment happened; the other covariates are the declared adjustment set, not a substitute for randomization.

```
star_unadjusted <- lm(readk ~ stark, data = star_df)
star_primary <- lm(readk ~ stark + lunchk + gender + ethnicity +
  experiencek + schoolidk, data = star_df)
```

On the common 3,735 primary rows, the unadjusted difference is 5.8955 and the adjusted coefficient is 6.6575. Adjustment can improve precision when the added variables predict reading. It is not needed to remove baseline confounding that a well-implemented randomization already prevents; this is Section 2.5's point about randomized groups, on a larger design. Adjustment also does not automatically preserve the causal target: when the covariates were recorded, which rows enter, and what the fitted comparison compares still need scrutiny.

The earlier 5.8151 comes from 3,745 rows, so it is not the same-row baseline for 6.6575; comparing the two would mix a change of records with a change of model. The same-row pair is 5.8955 against 6.6575. Similar unadjusted and adjusted coefficients are consistent with successful randomization; they do not show that implementation, missingness, or weighting problems are absent.

7.5 Clustered standard errors

Pupils are the outcome units. Schools are the randomization blocks and the approximately independent units for the uncertainty calculation; the schools themselves were not assigned to arms. Chapter 3, Section 2.3, showed that the uncertainty of a school-blocked comparison follows its within-school comparisons, and Figure 7 shows those comparisons for STAR.

In the 78 schools that observed pupils in both arms, the small-class advantage ranges from about -32 to 59 reading points, and it is negative in 25 schools. Pupils in one school share whatever made their school's difference large or small: its teachers, its intake, its classrooms. The 3,745 pupils therefore do not supply 3,745 independent pieces of information about the arm difference; the schools do. An average of 78 independent differences with SD 15.4192

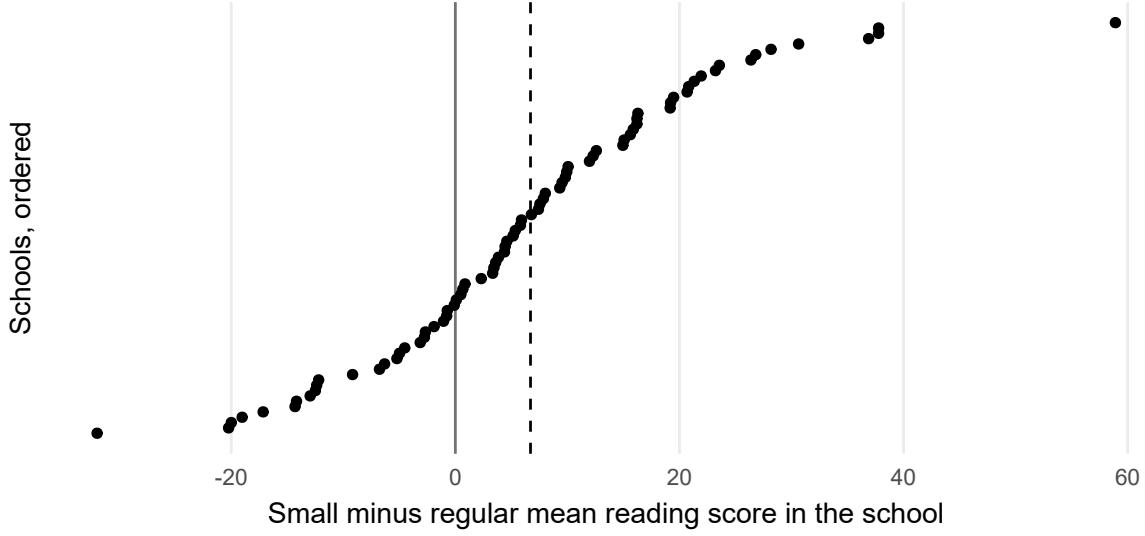


Figure 7: Small-minus-regular mean reading difference in each of the 78 schools that observed both class types, ordered by size. The dashed line is their unweighted mean, 6.7094.

has $\text{SE } 15.4192/\sqrt{78} = 1.7459$, close to the clustered standard errors below and far from the row-level 1.0377.

A **cluster-robust** covariance builds this into the robust covariance of Section 5.1: the design component stays outside, and only the middle changes. For school g , the score $\mathbf{s}_g = \mathbf{X}_g^\top \mathbf{e}_g$ sums the residual contributions **within** the school before an outer product is taken, and the CR1/HC1 covariance is

$$\widehat{\mathbf{V}}_{\text{CR1}} = c_G (\mathbf{X}^\top \mathbf{X})^{-1} \left(\sum_{g=1}^G \mathbf{s}_g \mathbf{s}_g^\top \right) (\mathbf{X}^\top \mathbf{X})^{-1}, \quad \mathbf{s}_g = \mathbf{X}_g^\top \mathbf{e}_g, \quad c_G = \frac{G}{G-1} \frac{n-1}{n-p}.$$

The within-school sums are what distinguish this from treating every pupil as independent, and c_G is a finite-cluster correction. With $G = 79$ schools, intervals use a t_{78} critical value, 1.9908 (Zeileis et al., 2020).

```
star_V <- sandwich::vcovCL(star_primary, cluster = star_df$schoolidk,
  type = "HC1", cadjust = TRUE)
sqrt(star_V["starksmall", "starksmall"])
```

```
[1] 1.697409
```

Table 6

Class-type comparisons and school-clustered intervals.

Model	<i>n</i>	Estimate	SE	Interval
Observed: unadjusted	3,745	5.8151	1.8476	[2.1368, 9.4935]
Primary: unadjusted	3,735	5.8955	1.8491	[2.2143, 9.5767]
Primary: adjusted	3,735	6.6575	1.6974	[3.2782, 10.0368]
Observed: school-adjusted	3,745	6.6273	1.7797	[3.0842, 10.1703]

Table 6 holds four distinct calculations, each with its own records and specification; each interval uses the school identifiers from that model’s own rows.

Set the clustered SE beside Section 5’s robust one. For the same 3,745-record unadjusted comparison, treating pupils as independent gives SE 1.0377, HC3 gives 1.0416, and school clustering gives 1.8476, nearly twice as large. Unequal spread across pupils was not this design’s problem; dependence within schools was, and no row-level robust covariance addresses it. The adjusted primary model shows the same pattern: row-level SE 0.9161, HC3 0.9509, clustered 1.6974.

These are cluster-asymptotic approximations, which improve as the number of schools grows; they are not exact randomization inference or a multilevel model. The design, not a residual plot, chooses the independence unit. And 79 clusters do not by themselves certify the approximation: the balance of information across schools and the influence of particular schools also matter.

7.6 Reporting the result and its limits

A chapter-authored report can now state:

On 3,735 complete two-arm kindergarten records, the common adjusted small-versus-regular class-type coefficient was 6.6575 reading points, with approximate school-clustered interval [3.2782, 10.0368]. Its causal mean-effect interpretation relies on documented within-school randomization, a verified link from recorded class type to original assignment, suitable complete-record selection, independent-school uncertainty, and a common-effect or target-matching weighting argument. It concerns assigned class type, one kindergarten reading outcome, and participating Tennessee schools, not realized headcount or a portable policy dose response.

Reading the report comes before judging it. In Chapter 1’s argument components, applied as in Chapter 8, Section 7, the sample audit and the model code carry **STAT-E** content; the design and covariance reasoning supplies **STAT-J**; the estimate and its clustered interval supply **STAT-I**; and the first sentence is the bounded **STAT-C** answer. **SUB-E** situates the pupils, the test, the cohort, and the school context; **SUB-J** adds the assignment record with the selection and weighting qualifications; **SUB-I** reads the coefficient as an assignment comparison only under that support; and **SUB-C** bounds the substantive claim. These are linked functions within one argument.

Judging the report separates what is *stated* from what is *demonstrated*. The randomization protocol is documented in the study’s own publications; the selection and weighting conditions and the link from recorded class type to original assignment are stated here, not shown. Where a necessary link cannot be assessed, the report names the limitation rather than assuming it away or quietly changing the target. The design literature discusses implementation and interpretation issues that one coefficient cannot settle (Chetty et al., 2011; Hanushek, 1999; Krueger, 1999; Nye et al., 2000), and the result does not transport by itself to other schools, periods, outcomes, or education systems. Those limits belong inside the answer; a crossed significance boundary is not what makes a result (Gelman et al., 2020; Gelman & Stern, 2006). Chapter 14 returns to dependence as a design fact, reuses this calculation as its example, and asks what a design supplies that adjustment cannot.

8 Reading Bridge

A factor enters a regression as indicators, and the one-way analysis of variance is that regression; a contrast’s uncertainty needs the covariance between coefficients, not the sum of two standard errors. The analysis of covariance adds a pretest: the posttest-only, gain-score, and adjusted analyses differ only in the pretest coefficient, and adjustment sharpens a randomized comparison without supplying what randomization supplies. Nested models on the same records and outcome are compared through their residual sums of squares, and information criteria rank models by penalized in-sample fit, never by held-out error. A squared predictor has no single slope, and a logged outcome asks a new question on a geometric-mean scale. HC3, the case bootstrap, and a clustered covariance each change the uncertainty for a stated reason and leave the coefficients alone; none repairs curvature, missing outcomes, weighting, or the absence of a design. In Project STAR, schools rather than pupils were the approximately independent units, and the standard error nearly doubled.

Every comparison in this chapter was the same at every value of the other predictors, which is what “additive” means, and Section 2.5’s common-slope check was the first test of that. Chapter 10 lets a comparison vary with another predictor and develops how to read, probe, and report such an interaction.

Key terms

Use this short reference to revisit the concepts introduced here. The Part One Concept Reference gives the broader reference and the principal home of each concept; indicator coding, partial slopes, coefficient covariance, and HC3 are explained in Chapters 7 and 8, and regression toward the mean in Chapter 7.

Table 7

Selected concepts for the regression toolbox.

Term and common alternatives	Meaning
ANCOVA (analysis of covariance)	A regression comparing groups while including a quantitative covariate with one common slope; its adjusted group means are read at a stated covariate value within common support. The posttest-only, gain-score and ANCOVA analyses are one regression whose pretest coefficient is 0, 1 or estimated; adjustment cannot supply what random assignment supplies.
Homogeneity of regression slopes (parallel slopes)	ANCOVA's condition that the outcome-covariate slope is the same in every group, checked by testing the group-by-covariate product block; when the slopes differ, the adjusted difference depends on the covariate value at which it is read.
Nested-model F test (extra sum of squares; block test)	Compares a reduced model with a fuller one fitted to the same records and outcome: the RSS reduction per added coefficient, divided by the fuller model's residual mean square, referred to F with the number of restrictions and the fuller model's residual df. It tests the added block jointly, not each coefficient separately.
Information criteria (AIC; BIC)	Penalized in-sample fit: $-2 \log \hat{L}$ plus $2k$ (AIC) or $k \log n$ (BIC), with k counting the coefficients and the residual variance. Smaller is preferred among models fitted to the same records and outcome scale; neither is held-out error.
Polynomial term (quadratic term; curvature)	A power of a predictor added as a further column so that the fitted mean can curve. The predictor's comparison then depends on its value; centering before squaring reduces collinearity without changing the fitted curve.
Log-transformed outcome (geometric-mean scale)	Fitting the logarithm of the outcome, so that a coefficient is an additive difference in expected log outcome and $100(e^\beta - 1)$ percent is a multiplicative difference in geometric means. A changed outcome scale is a changed question, and its criteria are not comparable with the original scale's.

Term and common alternatives	Meaning
Standardized coefficient (beta weight)	A partial slope rescaled by the predictor and outcome SDs, in outcome-SD units per predictor SD, with the other predictors held fixed. It differs from standardization over named profiles and does not measure causal importance.
Bootstrap (case resampling; percentile interval)	Refitting the model to many resamples of the records drawn with replacement and using the spread of the refitted estimates as an SE, or their middle 95% as a percentile interval. It treats the resampled units as independent and exchangeable; it does not repair dependence, thin support, measurement error, or an observational design.
Cluster-robust standard error (school-clustered covariance)	A coefficient covariance that sums residual contributions within clusters, such as schools, before combining them, allowing dependence within clusters under an approximation that treats clusters as independent. It changes uncertainty, not the coefficient, and does not repair missing outcomes or weighting.

Further reading

- The OpenIntro multiple-predictor and model-selection tutorial gives concrete indicator interpretations and criterion-based reading. We use its equation-first device without adopting automatic stepwise selection.
- The ISLR regression lab and the ISLR resampling lab connect fitting, factors, criteria, and resampling in one workflow (James et al., 2021).
- Fox (Fox, 2016) and Kutner and colleagues (Kutner et al., 2005) develop the linear-model algebra, the restriction statistic, and the diagnostics behind this chapter’s matrix section.
- Efron and Tibshirani (Efron & Tibshirani, 1993) and Davison and Hinkley (Davison & Hinkley, 1997) are the standard bootstrap references, including the bootstrap- t and BCa intervals named in Section 6.3.
- The sandwich documentation specifies the HC and CR covariance calls used here (Zeileis et al., 2020).

Exercises

Use the supplied numbers; rounding within 0.02 is acceptable unless a different precision is requested. These prompts do not require the appendices.

Exercise 1: Read indicator coefficients (Foundational)

A group-only regression of income (thousand dollars) on occupation type has observed group means 5.3741 (blue-collar, 44 records), 10.5595 (professional, 31), and 5.0523 (white-collar, 23). (1) With blue-collar as reference, give the intercept and the two indicator coefficients. (2) Give the intercept and both coefficients with white-collar as reference. (3) Explain why the fitted values are the same under both codings. (4) The full additive model's professional coefficient is 0.3242. Explain why it differs from the professional coefficient in (1) and why neither is "the effect of occupational type".

Exercise 2: A contrast needs covariance (Foundational)

In the full additive model the professional and white-collar coefficients are 0.3242 and 0.2353, with estimated variances 2.1406 and 0.9698 and covariance 0.9720, on 92 residual df. (1) The professional-minus-white-collar contrast is 0.0889. Calculate its variance, and explain why the covariance enters with a minus sign and what a positive covariance does to the result. (2) The contrast's SE is 1.08 and its 95% interval is $[-2.0561, 2.2338]$. A student instead adds the two coefficient SEs (1.4631 and 0.9848). State what that number is and why it is wrong. (3) Explain what the interval does and does not say about the two types.

Exercise 3: A nested comparison from residual sums of squares (Foundational)

On the same 98 records, the model with education and prestige has RSS 867.6537 on 95 df, and the full model adding the percentage of women and the two type indicators has RSS 630.8897 on 92 df. The 5% critical value of $F(3, 92)$ is 2.7036. (1) Calculate the F statistic for the added block. (2) State the decision at 5% and what it does and does not say about the three added columns. (3) Explain why the sequential sums of squares from a single `anova()` call on the full model would not answer this question.

Exercise 4: What information criteria compare (Foundational)

The full additive model has maximized log-likelihood -230.302 on 98 records, with AIC 474.6040 and BIC 492.6988; the model with education and prestige only has AIC 499.8328 and BIC 510.1727. (1) The criteria count seven fitted parameters for the full model. Say what the seven are, and why BIC charges the full model more for them than AIC does ($\log 98 = 4.585$). (2) Which model do the criteria favour, and on what condition is that comparison meaningful? (3) A log-income version of the full model has AIC 8.091. Explain why this number cannot be compared with 474.604.

Exercise 5: Percent and curvature (Integrative)

Two changed models of the occupation records raise different questions. (1) In a log-income model the prestige coefficient is 0.0207. Give the percent difference in geometric-mean income per prestige point and say what “geometric mean” adds to the reading. (2) In an income model with a squared prestige term, the fitted slope is 0.0697 thousand dollars per prestige point at prestige 30, 0.1665 at 50 and 0.2633 at 70, and the turning point is at prestige 15.5906; prestige ranges from 17.3 to 87.2. Explain why the model has no single prestige slope, and what the position of the turning point tells you. (3) State in one sentence what the curve claims about income and prestige over the observed range, and one reason that claim is a sensitivity check rather than a discovery.

Exercise 6: Choose the uncertainty calculation (Integrative)

In Project STAR the small-minus-regular kindergarten reading difference on 3,745 observed records is 5.8151 points. Three standard errors are available: 1.0377 treating pupils as independent, 1.0416 with HC3, and 1.8476 clustering pupils within their 79 schools. Treating pupils as independent gives the 95% interval $[3.7807, 7.8496]$. (1) Using $t_{.975,78} = 1.9908$, give the interval under the clustered calculation and compare its width with the first. (2) Say which interval a report should use and why, naming the unit at which assignment happened, and explain why HC3 barely changes the first standard error. (3) Name two problems that none of the three calculations repairs.

Exercise 7: One randomized study, three analyses (Integrative)

In the Baumann records, 66 pupils in three randomized instruction groups have a pretest and a posttest. With DRA as reference, the DRTA difference is 3.0909 (SE 0.9614) from the posttest alone, 3.8636 (SE 0.7779) from the gain score, and 3.6266 (SE 0.7362) from the analysis of covariance, whose pretest coefficient is 0.6932; the DRA and DRTA pretest means are 10.5000 and 9.7273. (1) Write the three analyses as one regression and give the pretest coefficient each uses. (2) Explain why all three estimate the same average difference under random assignment, and show how the pretest gap accounts for the differences between the three estimates. (3) Say which analysis you would report, and when that choice has to be made. (4) Suppose the groups had been intact classes that differed at the pretest. Explain what changes, naming the paradox that can arise.

Appendix A: Matrix Geometry, Restrictions, and Sandwich Constructions

This appendix derives the results that Sections 3.2, 5.2, and 7.5 used, including the HC3 estimator that Chapter 8, Appendix B, stated, and names the target least squares estimates when a linear mean is only an approximation. The exercises do not require it, and the Computer Lab carries the numerical reproductions.

Projection and the normal equations

Let $\mathbf{X}$ be $n \times p$ with full column rank, so $\mathbf{X}^\top \mathbf{X}$ is invertible. The least-squares criterion is $S(\boldsymbol{\beta}) = (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^\top (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})$. Expanding gives $S = \mathbf{y}^\top \mathbf{y} - 2\boldsymbol{\beta}^\top \mathbf{X}^\top \mathbf{y} + \boldsymbol{\beta}^\top \mathbf{X}^\top \mathbf{X} \boldsymbol{\beta}$, and differentiating with respect to $\boldsymbol{\beta}$ yields the normal equations

$$\mathbf{X}^\top \mathbf{X} \hat{\boldsymbol{\beta}} = \mathbf{X}^\top \mathbf{y}, \quad \text{equivalently} \quad \mathbf{X}^\top (\mathbf{y} - \mathbf{X} \hat{\boldsymbol{\beta}}) = \mathbf{0}.$$

The second form is the geometry: the residual vector is orthogonal to every column of $\mathbf{X}$. Hence $\hat{\mathbf{y}} = \mathbf{X} \hat{\boldsymbol{\beta}} = \mathbf{H} \mathbf{y}$ with $\mathbf{H} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$, and $\mathbf{H}$ is symmetric and idempotent, the orthogonal projection onto the column space of $\mathbf{X}$. Its trace is p , which is why the leverages h_{ii} sum to the number of fitted coefficients; for `m2` they sum to 6.

Two consequences used earlier follow immediately. Any full-rank recoding $\mathbf{X}\mathbf{A}$, with $\mathbf{A}$ invertible, spans the same column space, so $\mathbf{H}$ and $\hat{\mathbf{y}}$ are unchanged: releveling a factor or centering a predictor cannot alter a single fitted value. Adding a genuinely new column enlarges the space and can change every fitted value. And under the classical error model, $\hat{\mathbf{V}} = s^2(\mathbf{X}^\top \mathbf{X})^{-1}$ with $s^2 = RSS/(n - p)$, which is where every classical standard error in this chapter came from. Solving the normal equations directly for `m2` reproduces `lm()`'s coefficients to about 10^{-13} .

General linear restrictions

Let $\mathbf{R}$ be $q \times p$ of rank q and consider $\mathbf{R}\boldsymbol{\beta} = \mathbf{r}$. Since $\widehat{\boldsymbol{\beta}}$ has covariance $\sigma^2(\mathbf{X}^\top \mathbf{X})^{-1}$ under the classical model, the vector $\mathbf{R}\widehat{\boldsymbol{\beta}} - \mathbf{r}$ has covariance $\sigma^2 \mathbf{R}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{R}^\top$. Standardizing by that covariance and dividing by q gives the statistic of Section 3.2, which has an exact $F_{q, n-p}$ reference under Normal errors, because its numerator and denominator quadratic forms are independent chi-squares.

It is the same number the extra-sum-of-squares formula produces. Minimizing $S(\boldsymbol{\beta})$ subject to $\mathbf{R}\boldsymbol{\beta} = \mathbf{r}$ with a Lagrange multiplier gives the restricted estimator $\widetilde{\boldsymbol{\beta}} = \widehat{\boldsymbol{\beta}} - (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{R}^\top [\mathbf{R}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{R}^\top]^{-1} (\mathbf{R}\widehat{\boldsymbol{\beta}} - \mathbf{r})$, a correction applied only in the directions the restriction touches. Substituting it into the criterion shows that the increase in residual sum of squares is

$$RSS_R - RSS_F = (\mathbf{R}\widehat{\boldsymbol{\beta}} - \mathbf{r})^\top [\mathbf{R}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{R}^\top]^{-1} (\mathbf{R}\widehat{\boldsymbol{\beta}} - \mathbf{r}),$$

which is exactly the numerator of the restriction statistic. Fitting the reduced model and differencing, or evaluating the quadratic form in the unrestricted fit, are two routes to one number; for the two type coefficients of `m2` both return 0.0319. Replacing $s^2(\mathbf{X}^\top \mathbf{X})^{-1}$ by a robust covariance in the middle of the expression gives the Wald statistic of Section 5.2. The algebra survives the substitution; the exact reference distribution does not.

The HC3 sandwich

Chapter 8, Appendix B, stated the HC3 estimator; this subsection shows where it comes from. Write the coefficient error as $\widehat{\boldsymbol{\beta}} - \boldsymbol{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \boldsymbol{\varepsilon}$. Under independent rows with possibly different variances σ_i^2 , its exact covariance is

$$(\mathbf{X}^\top \mathbf{X})^{-1} \left(\sum_i \sigma_i^2 \mathbf{x}_i \mathbf{x}_i^\top \right) (\mathbf{X}^\top \mathbf{X})^{-1},$$

a *sandwich* with the design component as bread and the error-scale component as filling. The classical estimator substitutes a common s^2 , which collapses the filling to $s^2 \mathbf{X}^\top \mathbf{X}$. The HC family substitutes each row's own squared residual instead. Because $E(e_i^2) = \sigma_i^2(1 - h_{ii})$ under the classical model, the raw residual understates the error at high-leverage rows; HC3 corrects by dividing by $(1 - h_{ii})^2$, which is deliberately slightly conservative and performs well in small samples (Long & Ervin, 2000; MacKinnon & White, 1985). Built directly from these pieces, the estimator reproduces `sandwich::vcovHC(m2, type = "HC3")` to about 10^{-12} .

The CR1 cluster sandwich

Cluster robustness changes the filling, not the bread. Partition the rows into G clusters and let $\mathbf{X}_g$ and $\mathbf{e}_g$ hold cluster g 's rows and residuals. The cluster score $\mathbf{s}_g = \mathbf{X}_g^\top \mathbf{e}_g$ sums the within-cluster contributions *before* the outer product is taken, so the estimator of Section 7.5 retains the within-cluster residual correlations that the independent-row filling discards. If every cluster held one row, the filling would reduce to the HC0 form, which is why a row-level robust SE is the special case that assumes away the dependence.

The asymptotics are in G , not in n : 3,745 pupils in 79 schools supply 79 approximately independent pieces of information for this covariance, which is why the clustered SE of 1.8476 is so much larger than the row-level 1.0377 and why a t_{G-1} reference is used. For the observed-reading model the finite-cluster factor is $c_G = 1.0131$, and building the estimator from the school scores reproduces `sandwich::vcovCL(..., type = "HC1", cadjust = TRUE)` to about 10^{-13} .

What least squares estimates when the mean is not linear

A linear regression is often fitted where the conditional mean of the outcome is not exactly linear in the predictors, as Section 4.1's curved prestige component shows. Least squares then still estimates a defined target: the **best linear projection**, the coefficients that minimize expected squared error over the population the records represent,

$$(\beta_0^*, \beta_1^*) = \arg \min_{a,b} E\{(Y - a - bX)^2\}, \quad \beta_1^* = \frac{\text{Cov}(X, Y)}{\text{Var}(X)},$$

$$\beta^* = \{E(\mathbf{x}\mathbf{x}^\top)\}^{-1} E(\mathbf{x}Y).$$

With one predictor the target slope is the population covariance over the predictor's variance, the population version of Chapter 7's covariance-to-slope identity; with several, it is the population version of the normal equations. The target is well defined, but it is an average: the projection slope of income on prestige in the additive model averages the curve's steeper and flatter stretches, weighted by where the occupations' prestige values fall. Its value therefore depends on the distribution of the predictors, and a sample with a different prestige mix would have a different target even with the same curve. It is a declared alternative target, not a rescue for a misspecified mean. HC3 standard errors remain appropriate for it under independent sampling, because the curvature the linear fit misses behaves like one more source of unequal spread (Fox, 2016).

Appendix B: Bootstrap Schemes and Intervals

Section 6 used case resampling and said why. Table 8 sets it beside the other two schemes on the same model, the same three targets, the same seed, and $B = 2000$.

Table 8

Three bootstrap schemes for the full occupation model, with each scheme’s bootstrap SE for three targets (seed 90909, $B = 2000$).

Scheme	What is resampled	Assumption carried	Education slope	prof – wc	prof coefficient
Case	Whole records	Records independent and exchangeable	0.3699	1.1187	1.9926
Residual	Rescaled residuals, moved across records; design fixed	One error distribution for every record	0.2820	1.1081	1.4861
Wild	Each record’s own rescaled residual, sign flipped at random; design fixed	Each record keeps its own error scale	0.3673	1.1151	1.9767

Both design-fixed schemes rebuild outcomes from the fitted values. Residual resampling draws leverage-rescaled residuals $e_i/\sqrt{1-h_{ii}}$ with replacement and adds them to the fitted values, which treats the rescaled residuals as exchangeable across rows. The wild bootstrap keeps each row’s own rescaled residual and multiplies it by ± 1 with equal probability, so each row retains its own error scale.

Read the table by column. For the education slope the residual scheme is the outlier, 0.2820, close to the classical 0.2752, as expected when both assume one common error distribution. The wild bootstrap, which does not, lands at 0.3673, near the case result and HC3’s 0.3823. For the professional coefficient the pattern is stronger: 1.9926, 1.4861, and 1.9767. When the schemes disagree, the disagreement is about the error model, and it is a reason to say which scheme was used.

Section 6.3 defined the percentile interval, the middle 95% of the resampling distribution, and the normal-approximation interval, which uses that distribution only for its spread and so ignores its skewness. Two higher-order intervals are worth describing in words.

- The **bootstrap- t interval** resamples a studentized quantity, the resampled estimate minus the original estimate, divided by the resample’s own standard error, and inverts the empirical quantiles of that ratio. It needs a standard error inside every resample, and it is more accurate when the studentized quantity is closer to pivotal than the estimate itself.

- The **BCa interval** adjusts the percentile endpoints for median bias and for an acceleration constant that captures how the standard error changes with the parameter. It is the usual recommendation when the resampling distribution is visibly skewed (Davison & Hinkley, 1997; Efron & Tibshirani, 1993).

For the three targets of Section 6 the skewness values -0.3891 , 0.2635 , and 0.3785 are mild, so percentile and normal-approximation endpoints agree closely and the higher-order corrections would change little. That agreement is a property of this example, not a general licence.

References

- Akaike, H. (1974). A new look at the statistical model identification. *IEEE Transactions on Automatic Control*, 19(6), 716–723. <https://doi.org/10.1109/TAC.1974.1100705>
- Baumann, J. F., Seifert-Kessell, N., & Jones, L. A. (1992). Effect of think-aloud instruction on elementary students’ comprehension monitoring abilities. *Journal of Reading Behavior*, 24(2), 143–172. <https://doi.org/10.1080/10862969209547770>
- Chetty, R., Friedman, J. N., Hilger, N., Saez, E., Schanzenbach, D. W., & Yagan, D. (2011). How does your kindergarten classroom affect your earnings? Evidence from Project STAR. *The Quarterly Journal of Economics*, 126(4), 1593–1660.
- Cohen, J., Cohen, P., West, S. G., & Aiken, L. S. (2003). *Applied multiple regression/correlation analysis for the behavioral sciences* (3rd ed.). Lawrence Erlbaum Associates.
- Davison, A. C., & Hinkley, D. V. (1997). *Bootstrap methods and their application*. Cambridge University Press. <https://doi.org/10.1017/CBO9780511802843>
- Efron, B. (1979). Bootstrap methods: Another look at the jackknife. *The Annals of Statistics*, 7(1), 1–26. <https://doi.org/10.1214/aos/1176344552>
- Efron, B., & Tibshirani, R. J. (1993). *An introduction to the bootstrap*. Chapman and Hall/CRC. <https://doi.org/10.1201/9780429246593>
- Finn, J. D., & Achilles, C. M. (1990). Answers and questions about class size: A statewide experiment. *American Educational Research Journal*, 27(3), 557–577.
- Fox, J. (2016). *Applied regression analysis and generalized linear models* (3rd ed.). SAGE Publications.
- Fox, J., & Weisberg, S. (2019). *An R companion to applied regression* (3rd ed.). SAGE Publications. <https://www.john-fox.ca/Companion/index.html>
- Fox, J., Weisberg, S., & Price, B. (2022). *carData: Companion to applied regression data sets* (Versions 3.0-5). Comprehensive R Archive Network. <https://doi.org/10.32614/CRAN.package.carData>
- Freedman, D. A. (1981). Bootstrapping regression models. *The Annals of Statistics*, 9(6), 1218–1228. <https://doi.org/10.1214/aos/1176345638>
- Gelman, A., Hill, J., & Vehtari, A. (2020). *Regression and other stories*. Cambridge University Press. <https://doi.org/10.1017/9781139161879>

- Gelman, A., & Stern, H. (2006). The difference between “significant” and “not significant” is not itself statistically significant. *The American Statistician*, 60(4), 328–331. <https://doi.org/10.1198/000313006X152649>
- Hanushek, E. A. (1999). Some findings from an independent investigation of the Tennessee STAR experiment and from other investigations of class size effects. *Educational Evaluation and Policy Analysis*, 21(2), 143–163.
- James, G., Witten, D., Hastie, T., & Tibshirani, R. (2021). *An introduction to statistical learning: With applications in R* (2nd ed.). Springer. <https://doi.org/10.1007/978-1-0716-1418-1>
- Kleibers, C., & Zeileis, A. (2008). *Applied econometrics with R*. Springer. <https://doi.org/10.1007/978-0-387-77318-6>
- Krueger, A. B. (1999). Experimental estimates of education production functions. *The Quarterly Journal of Economics*, 114(2), 497–532.
- Kutner, M. H., Nachtsheim, C. J., Neter, J., & Li, W. (2005). *Applied linear statistical models* (5th ed.). McGraw-Hill/Irwin.
- Long, J. S., & Ervin, L. H. (2000). Using heteroscedasticity consistent standard errors in the linear regression model. *The American Statistician*, 54(3), 217–224. <https://doi.org/10.1080/00031305.2000.10474549>
- Lord, F. M. (1967). A paradox in the interpretation of group comparisons. *Psychological Bulletin*, 68(5), 304–305. <https://doi.org/10.1037/h0025105>
- MacKinnon, J. G., & White, H. (1985). Some heteroskedasticity-consistent covariance matrix estimators with improved finite sample properties. *Journal of Econometrics*, 29(3), 305–325. [https://doi.org/10.1016/0304-4076\(85\)90158-7](https://doi.org/10.1016/0304-4076(85)90158-7)
- Mosteller, F. (1995). The Tennessee study of class size in the early school grades. *The Future of Children*, 5(2), 113–127.
- Nye, B., Hedges, L. V., & Konstantopoulos, S. (2000). The effects of small classes on academic achievement: The results of the Tennessee class size experiment. *American Educational Research Journal*, 37(1), 123–151.
- Schwarz, G. (1978). Estimating the dimension of a model. *The Annals of Statistics*, 6(2), 461–464. <https://doi.org/10.1214/aos/1176344136>
- Stock, J. H., & Watson, M. W. (2007). *Introduction to econometrics* (2nd ed.). Addison Wesley.
- White, H. (1980). A heteroskedasticity-consistent covariance matrix estimator and a direct test for heteroskedasticity. *Econometrica*, 48(4), 817–838. <https://doi.org/10.2307/1912934>
- Zeileis, A. (2004). Econometric computing with HC and HAC covariance matrix estimators. *Journal of Statistical Software*, 11(10), 1–17. <https://doi.org/10.18637/jss.v011.i10>
- Zeileis, A., Köll, S., & Graham, N. (2020). Various versatile variances: An object-oriented implementation of clustered covariances in R. *Journal of Statistical Software*, 95(1), 1–36. <https://doi.org/10.18637/jss.v095.i01>